

Project Assignment 4: Synthesis

Design Thinking | Student ID: s-01141 | August 11, 2026

DESIGN CHALLENGE

Redesign the online learning experience for first-generation university students in a world in which access to high-speed internet is unreliable.

This synthesis moves from interview evidence to interpretation, insight, Point of View, and a focused design opportunity.

1 Task 1: Interpreting and Abducting

1.1 Quotation Flowers

1.1.1 Quotation 1: The Silent Trade-Off

“ I’ve started screenshotting the slides and reading them like a textbook. It’s not the same- I miss the explanations, but it’s the only way I can keep up. ”

— Maria, The Rural Student

💡 Interpretations

- 💡 **Creative workaround** - She’s found a way to survive, but it’s not real learning.
- ⚖️ **Trade-off** - Accepting a degraded experience because the alternative is worse.
- 😞 **Shame** - She knows she’s doing something “not the same.”
- ❓ **Worry** - “Will my degree mean the same thing?”
- 🕒 **Unseen effort** - She’s working harder, not smarter.
- 🛠️ **Survival mindset** - Getting through rather than truly understanding.

Inference & Extrapolation

Maria is making a fundamental trade-off that she knows is suboptimal. She has accepted a degraded learning experience because the alternative, not learning at all, is worse. This suggests she values her education deeply but feels the system is not designed for her. There is an underlying sense of shame: she is doing something she knows is “not the same” and worries that the difference will show in her performance. She is spending more time on her studies than her peers, but getting less out of them. This is unsustainable and risks burnout or disengagement.

1.1.2 Quotation 2: The Impossible Choice

“ My professors keep saying “just watch the recording later,” but they don’t understand, downloading a 45-minute video uses up half my data allowance for the month. ”

— Maria, The Rural Student

💡 Interpretations

- ⚠️ **Disconnect** - Professors assume access that does not exist.
- ✖️ **Impossible choice** - Education versus basic necessities.
- 🚫 **Exclusion** - The system is designed for someone else.
- 🔇 **Silence** - She may have stopped explaining her situation.
- 😞 **Frustration** - The “solution” creates a worse problem.

Inference & Extrapolation

This reveals a fundamental disconnect between what professors assume is accessible and what is actually possible. Maria is faced with an impossible choice: use her limited data

for a lecture or preserve it for essential communication, job applications, or emergencies. She may have stopped explaining her situation to professors because she has learned that explaining does not necessarily lead to understanding or accommodation. This silence is dangerous because it means problems can remain invisible, unreported, and unaddressed.

1.1.3 Quotation 3: The Burden of Apology

“ I feel like I’m always apologizing to my professors for late work, to my group members for not being able to join meetings, to my family for using too much data. ”

— James, The Sharing Student

💡 Interpretations

- 🔒 **Internalized blame** - He sees himself as the problem.
- 🕒 **Burden** - He feels like an inconvenience.
- ☹️ **Corrosive shame** - It affects his sense of belonging.
- ⚖️ **Structural inequity** - He is apologizing for circumstances beyond his control.
- 👥 **Belonging** - He may no longer feel that he belongs.

Inference & Extrapolation

James is internalizing a narrative that he is the problem. He is apologizing for circumstances beyond his control, structural inequities that the university has not adequately addressed. This constant apology is corrosive to his sense of belonging and self-worth. He may be starting to believe that he does not belong at university. His experience of apology may also affect his academic identity, making him less likely to speak up in class, ask for help, or persist through difficulties.

1.1.4 Quotation 4: The Double Bind

“ I want to ask for help, but I don’t even know what to ask. I don’t know what I don’t know. ”

— James, The Sharing Student

💡 Interpretations

- 🔒 **Double bind** - He needs help but lacks the knowledge required to ask for it.
- 📋 **Hidden curriculum** - Unwritten rules first-generation students may not inherit.
- ⏸️ **Paralysis** - He is stuck between needing help and not knowing how to seek it.
- ❓ **Misunderstood** - He may appear disengaged when he is actually struggling.
- 🔧 **Skills gap** - Institutional navigation skills are missing.

Inference & Extrapolation

This reveals the compounded disadvantage of being a first-generation student. Not only

may James lack family guidance about university; he may also lack the vocabulary and confidence required to seek institutional support.

He is caught in a double bind: he needs help, but asking for help requires some of the very institutional knowledge he lacks. Professors and peers may interpret his behaviour as a lack of effort or interest, when in reality he may be paralyzed by uncertainty about how to navigate the system.

Without intervention, he may never develop the confidence or knowledge needed to advocate effectively for himself.

1.1.5 Quotation 5: The Invisible Student

“ I don't think my professors realize that some of us are working and studying at the same time. They think we're all just students with nothing else to do. ”

— Aisha, The Working Student

💡 Interpretations

👁️ **Invisibility** - Her reality is not recognized.

🎓 **Assumed typical student** - The system appears designed around someone else.

👊 **Resentment** - Frustration with the system is growing.

⚖️ **Compromise** - She risks doing neither work nor study as effectively as she could.

💀 **Risk** - Persistent strain can increase disengagement and dropout risk.

Inference & Extrapolation

Aisha feels invisible because her reality is not recognized by the institution. There is an assumed “typical student” who has enough time and flexibility to organize life around university expectations, and she does not fit that model.

She is navigating two demanding worlds with limited support from either. Her statement that “professors don't realize” suggests that the problem is not simply workload; it is also a lack of recognition. If this continues, frustration may become emotional disengagement, especially if she concludes that the institution is unwilling to accommodate students whose lives do not fit the assumed norm.

1.2 Key Insights Summary

Emerging Themes from the Data

- 💡 **Students develop creative workarounds.** These strategies help them survive, but they are often exhausting, inefficient, and unsustainable.
- ⚠️ **The system assumes access that does not exist.** This creates impossible choices between education, communication, work, and basic needs.
- 😞 **Shame and apology shape the experience.** Students can begin to interpret structural barriers as personal failures.
- 📖 **First-generation students may lack institutional knowledge.** This compounds technical barriers because even seeking help can require knowledge they have not yet acquired.
- 👁️ **Students feel invisible.** Their realities are not always recognized by systems designed around an assumed “typical” student.

2 Task 2: Formulate a Point of View

2.1 Point of View Statement 1: Maria’s Story

👤 I met

Maria, a resourceful and determined rural student who drives 20 minutes to the library just to access Wi-Fi for her lectures, and who has learned to screenshot slides to save her precious data.

🔍 I was inspired to discover

that she knows she is not learning as deeply as her peers and worries constantly that her degree will not mean the same thing, yet she persists because she values her education more than the system values her presence in it.

➔ It’s all about helping Maria

learn effectively and confidently without constantly having to make painful trade-offs between her education and her limited resources.

2.2 Point of View Statement 2: James’s Story

👤 I met

James, a first-generation student who has learned to apologize, for late work, for missed meetings, and for needing help, and who has internalized the belief that he is a burden to the system.

🔍 I was inspired to discover

that he desperately wants to ask for help but does not even know what to ask, because the institutional knowledge that some of his peers inherited from their families is a foreign language to him.

➔ It’s all about helping James

navigate and belong in the university environment without feeling like an outsider who has to apologize for existing.

2.3 Point of View Statement 3: Aisha's Story

I met

Aisha, a working student who juggles a full-time job and her studies, trying to fit learning into the cracks of her day and feeling like her professors do not even know she exists.

I was inspired to discover

that she feels invisible to the academic community, which seems to assume that students have no other responsibilities and can structure their time around university expectations.

It's all about helping Aisha

learn on her own terms, with flexibility and recognition that her reality is different from the "typical student" the system was designed for.

3 Choosing a Direction

Which PoV appeals to me most?

PoV Statement 2 - James's Story

Why?

James's experience of "always apologizing" captures something that goes beyond technical connectivity issues. It speaks to the *emotional and social dimensions* of being a first-generation student in an online environment: the shame, the invisibility, and the sense of being a burden.

The problem here is therefore not simply unreliable internet. It is that the broader learning environment can be designed in ways that make students like James feel they do not belong.

This framing opens up the most interesting design space. Solutions would not just be technical. They could involve rethinking how we build community, how we communicate expectations, and how we make institutional knowledge accessible.

The opportunity is ultimately about **belonging**, not just connectivity.

The problem I want to pursue:

How might we help first-generation students navigate online learning without feeling like outsiders who have to constantly apologize for their circumstances?

Submission Checklist

- ✓ **5 Quotation Flowers** with surrounding interpretations and extrapolations.
- ✓ **Key insights** extracted from the interpretation process.
- ✓ **3 Point of View statements** using the PoV template.

- ☑ **Selection of PoV 2** as the most promising direction for the solution phase.

This assignment demonstrates the application of synthesis in Design Thinking, moving from raw data to interpretation, insight, and creative reframing of the design challenge.
